5. There are two sequences (y_n) and (z_n) such that $0 \leq y_n \leq z_n$ for all n. The sequence (z_n) is convergent and its limit is zero. Prove that the sequence (y_n) is convergent to zero.

Proof: Given, the sequence (z_n) is convergent and its limit is 0.

$\therefore \forall \varepsilon \in \mathbb{R}, \varepsilon > 0, \exists N_1 \in \mathbb{N}$ s.t. $\forall n \in \mathbb{N}, n > N_1, |z_n - 0| < \varepsilon$

$\Rightarrow -\varepsilon < z_n < \varepsilon$. We know $\forall n, 0 \leq y_n \leq z_n$

$\therefore \boxed{-\varepsilon < 0 \leq y_n \leq z_n < \varepsilon}$ for $\forall n \in \mathbb{N}, n > N_1$

$\therefore -\varepsilon < y_n < \varepsilon$ for $\forall n \in \mathbb{N}, n > N_1$. $\Rightarrow |y_n - 0| < \varepsilon \ \forall n \in \mathbb{N}, n > N_1$

$\boxed{\therefore \text{The sequence } y_n \text{ is convergent to } 0}$ $\square$